

Exploded View

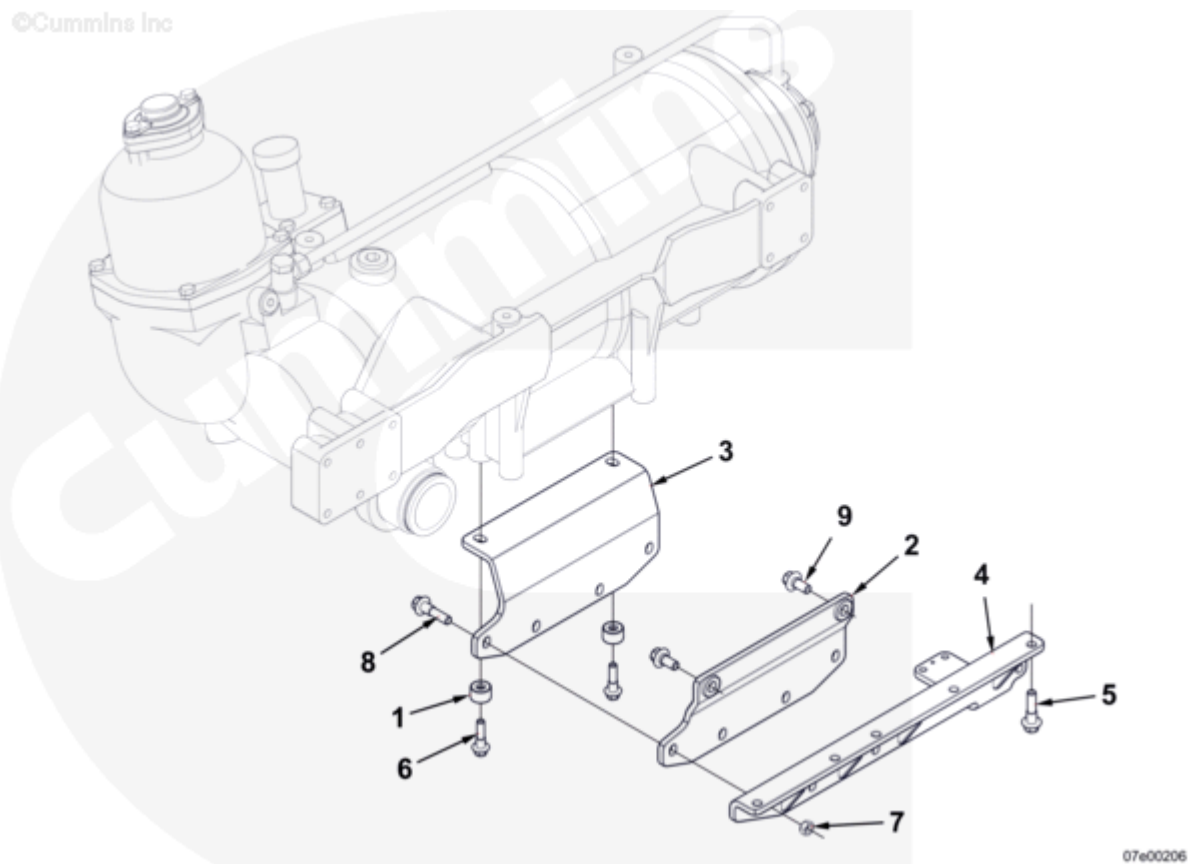


Figure 1 - Three-Piece Mounting Bracket Exploded View

1. Spacer
2. Bracket, Middle
3. Bracket, Front
4. Bracket, Rear
5. Screw
6. Screw
7. Nut
8. Screw
9. Screw.

Remove

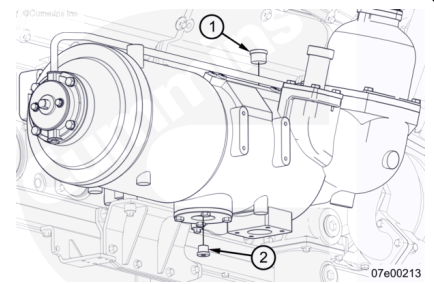
Single-piece Bracket

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Place a suitable container under the drain plug (2).

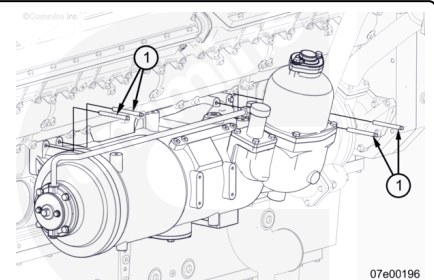
Remove the plug and let the oil drain.

Note : Remove the vent plug (1) to help empty the filter.

Install the drain plug (2) and vent plug (1) in the filter body.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four mounting capscrews from Eliminator™ filter head at indicated locations.

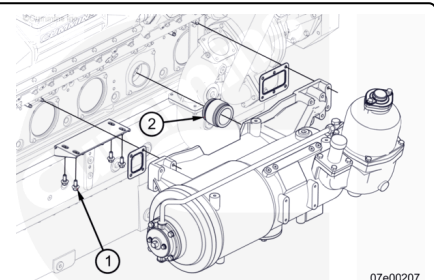
Install the four guide studs (1) and remove the six remaining capscrews.

Remove the four support capscrews (1).

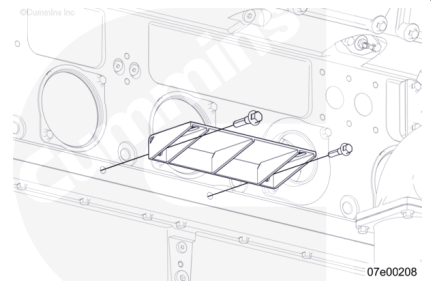
Remove the filter over the guide studs while removing the connection tube (2) from the hand hole cover.

Remove the gaskets over the guide studs and discard the gaskets.

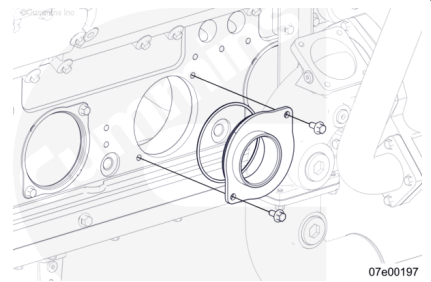
Remove the four guide studs from the engine block.



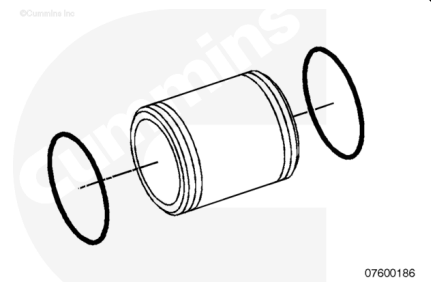
Remove the bracket mounting capscrews and the bracket.



Remove the Eliminator™ handhole cover and discard the handhole cover o-ring.
([/qs3/pubsys2/xml/en/bulletin/3666253.html](http://qs3/pubsys2/xml/en/bulletin/3666253.html))



Remove and discard the connection tube o-rings.



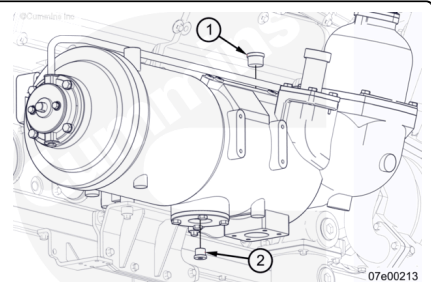
Three-piece Bracket

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Place a suitable container under the drain plug (2).

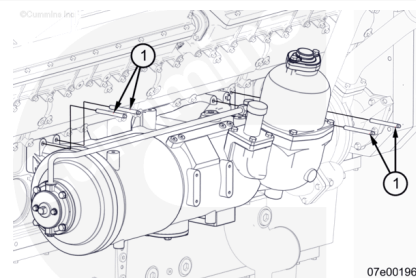
Remove the plug and let the oil drain.

Note : Removing the vent plug (1) helps to empty the filter.

Install the drain plug (2) and vent plug (1) in the filter body.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the four mounting capscrews from Eliminator™ filter head at indicated locations.

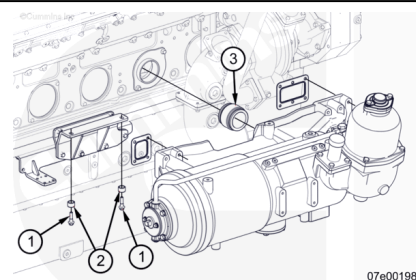
Install the four guide studs (1) and remove the six remaining capscrews.

Remove the two support capscrews (1) and spacers (2).

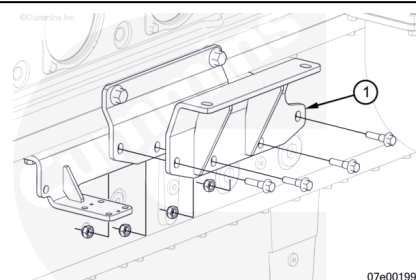
Remove the filter over the guide studs while removing the connection tube (3) from the hand hole cover.

Remove the gaskets over the guide studs and discard the gaskets.

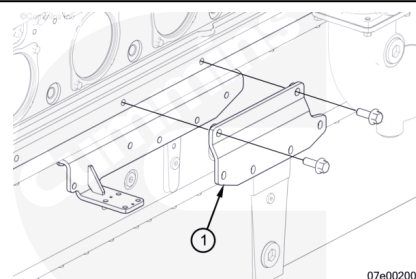
Remove the four guide studs from the engine block.



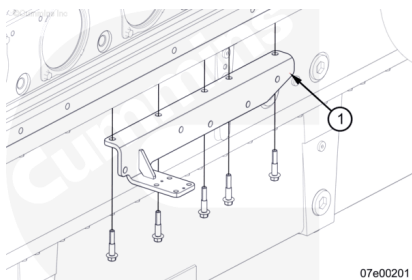
Remove the front bracket mounting capscrews with nuts and the front bracket (1).



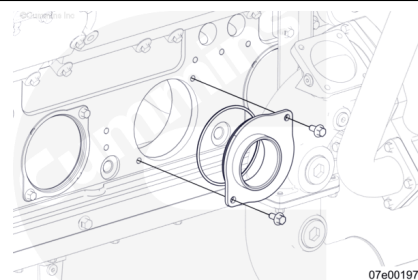
Remove the middle bracket mounting capscrews and the middle bracket (1).



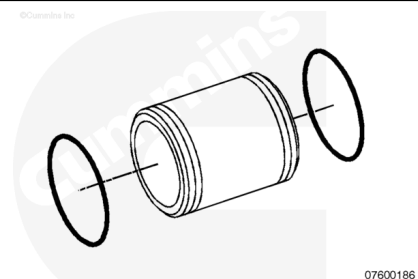
Remove the rear bracket mounting capscrews and the rear bracket (1).



Remove the Eliminator™ filter handhole cover and gasket.



Remove and discard the connection tube o-rings.



Disassemble

For a detailed Disassemble procedure, use the Maintenance Section in the Eliminator™ Filter, Bulletin 3666253. (</qs3/pubsys2/xml/en/bulletin/3666253.html>)



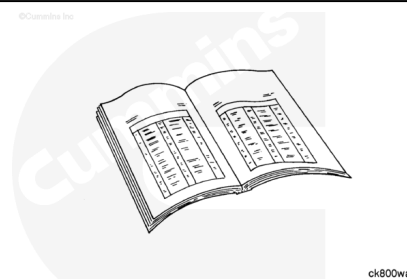
Clean and Inspect for Reuse

For a detailed Clean and Inspect for Reuse procedure, use the Maintenance Section in the Eliminator™ Filter, Bulletin 3666253. (</qs3/pubsys2/xml/en/bulletin/3666253.html>)



Assemble

For a detailed Assemble procedure, use the Maintenance Section in the Eliminator™ Filter, Bulletin 3666253. (</qs3/pubsys2/xml/en/bulletin/3666253.html>)

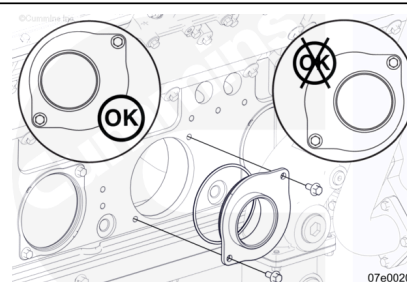


Install

Single-piece Bracket

⚠ CAUTION ⚠

The Eliminator™ filter can be installed on either side of the engine. Check to be sure the lubricating oil transfer tube has been installed correctly. The filter must be on the opposite bank from the oil transfer tube. If the Eliminator™ filter is installed on the wrong bank, the engine will be severely damaged because of the lack of lubrication.



Note : It is recommended that you install the eliminator with the three-piece bracket, where applicable.

Install the Eliminator™ filter handhole cover with the handhole cover o-ring.

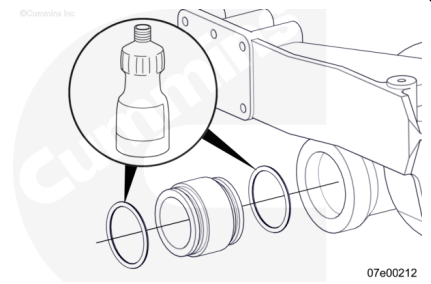
Make sure the handhole cover is in correct position on engine.

Tighten the handhole cover capscrews.

Torque Value: 45 n•m [33 ft-lb]

Lubricate the new o-ring seals with vegetable oil and install on both ends of the connection tube.

Install the connection tube into the filter housing.



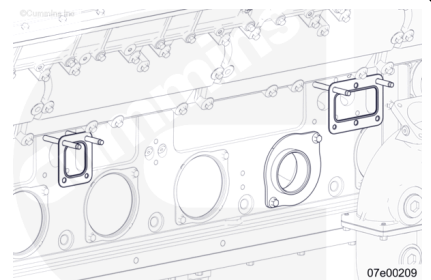
Install four guide studs to the engine in the locations shown.

Note : Do not use threaded guide studs. This will aid in guide stud removal once the Eliminator™ filter is installed.

Install new gaskets over the guide studs.

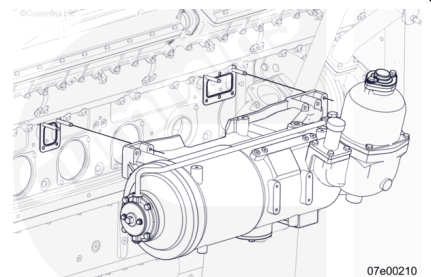
Make sure the six mounting holes are visible through each gasket.

Note : The gasket part number must face away from the cylinder block.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the filter over the guide studs while guiding the connection tube into the handhole cover.

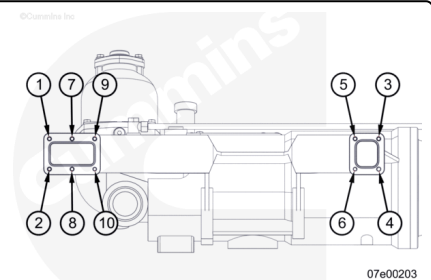
Install six capscrews and washers and hand-tighten.

Remove the guide studs and install the remaining capscrews and washers, and hand-tighten.

Tighten the capscrews in sequence illustrated.

Torque Value: 45 n•m [33 ft-lb]

Note : Due to the difficulty in accessing the lower capscrews, the use of a 3/8 drive socket and torque wrench is recommended to gain access to achieve the correct torque value on these capscrews.



Install the support bracket to the engine and Eliminator™. Hand-tighten all of the capscrews to align the bracket with the block and the Eliminator™ filter.

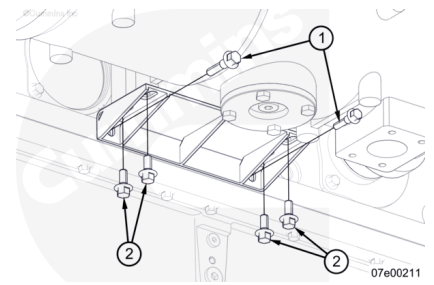
Tighten the capscrews.

Torque Value:

Pan Rail (1) 80 n•m [59 ft-lb]

Torque Value:

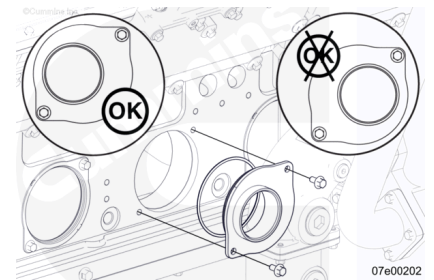
Eliminator™ Filter (2) 68 n•m [50 ft-lb]



Three-piece Bracket

⚠ CAUTION ⚠

The Eliminator™ filter can be installed on either side of the engine. Check to be sure the lubricating oil transfer tube has been installed correctly. The filter must be on the opposite bank from the oil transfer tube. If the Eliminator™ filter is installed on the wrong bank, the engine will be severely damaged because of the lack of lubrication.



Install the Eliminator™ filter handhole cover with a new gasket.

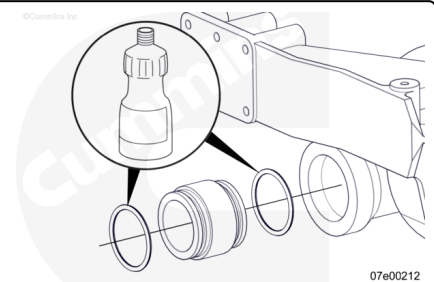
Make sure handhole cover is in correct position on engine.

Tighten the capscrews in handhole cover.

Torque Value: 45 n•m [33 ft-lb]

Lubricate the new o-ring seals with vegetable oil and install on both ends of the connection) tube.

Install the connection tube into the filter housing.

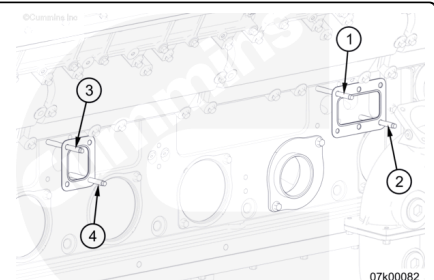


Install four guide studs to the engine in locations illustrated.

Note : Do not use threaded guide studs. This will aid in guide stud removal once the Eliminator™ filter is installed.

Install new gaskets over the guide studs.

Make sure the six mounting holes are visible through each gasket.

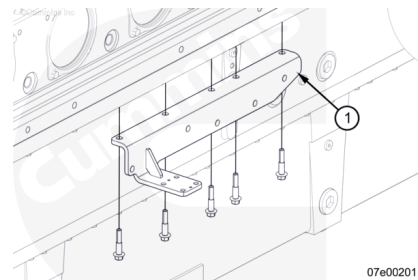


Note : The gasket part number must face away from the cylinder block.

Use the mounting capscrews to install the rear bracket (1) to the oil pan.

Tighten the capscrews.

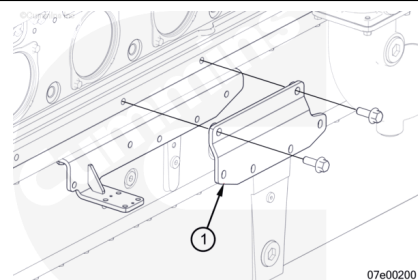
Torque Value: 45 n•m [33 ft-lb]



07e00201

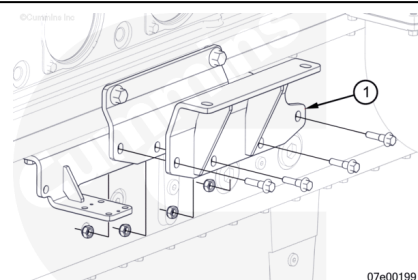
Use the bracket mounting capscrews to install the middle bracket (1). Tighten the capscrews.

Torque Value: 80 n•m [59 ft-lb]



07e00200

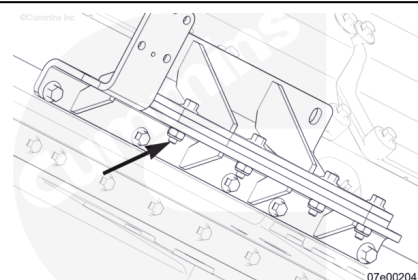
Use the bracket capscrews and nuts to install the front bracket (1).



07e00199

The capscrews should go through all three brackets.

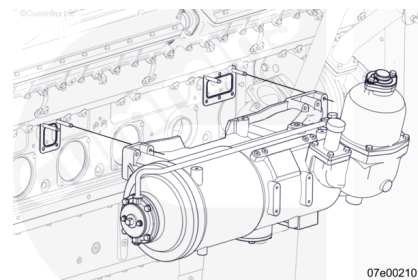
Do **not** tighten the capscrews and nuts.



07e00204

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07e00210

Install the filter over the guide studs while guiding the connection tube into the handhole cover.

If the installation requires spacers, install spacer with new gaskets between filter housing and the cylinder block.

Install new gaskets between the filter body and cylinder block.

Note : The front bottom capscrew needs to be inserted before the Eliminator™ filter is moved into place, due to limited clearance.

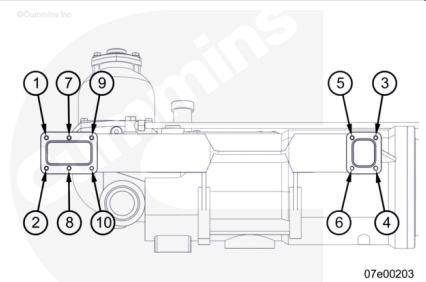
Install six capscrews and washers and hand-tighten.

Remove the guide studs and install the remaining capscrews and washers, hand-tighten to the eliminator manifold.

Tighten the capscrews in sequence illustrated.

Torque Value: 45 n•m [33 ft-lb]

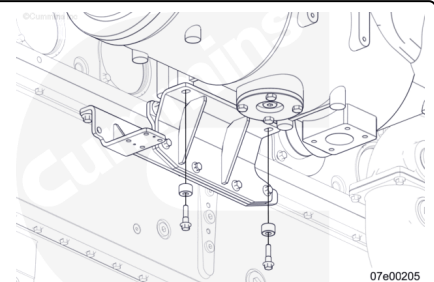
Note : Due to the difficulty in accessing the lower capscrews, the use of a 3/8 drive socket and torque wrench is recommended to gain access to achieve the correct torque value on these capscrews.



Use capscrews and spacers to secure the filter head front support bracket to the bottom of the filter head.

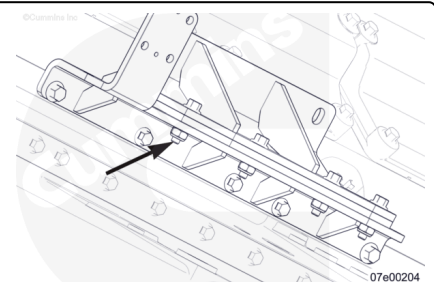
Install and tighten the two lower support capscrews.

Torque Value: 45 n•m [33 ft-lb]



Tighten the capscrews going through all three support brackets.

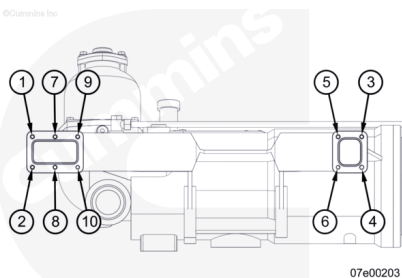
Torque Value: 45 n•m [33 ft-lb]



Loosen capscrews in sequence illustrated by at least one full turn.

Tighten capscrews in sequence illustrated in order from 1 to 10.

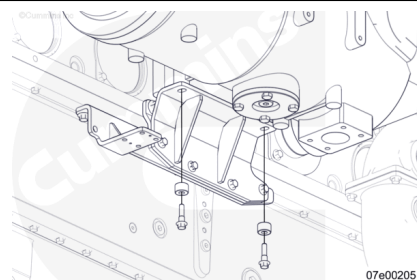
Torque Value: 45 n•m [33 ft-lb]



Loosen capscrews by at least one full turn.

Tighten both capscrews again.

Torque Value: 45 n·m [33 ft-lb]

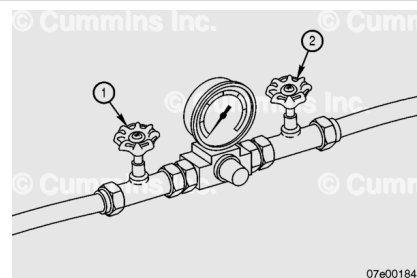


Pressure Test

This test will indicate oil pressure before and after the oil flows through the Eliminator™ filter.

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves (1 and 2) to eliminate gauge error. Use a gauge with a minimum pressure capacity of 1400 kPa [200 psi].

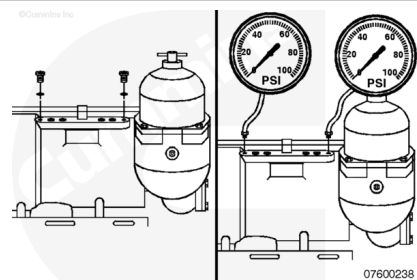
Be sure the engine is switched OFF.



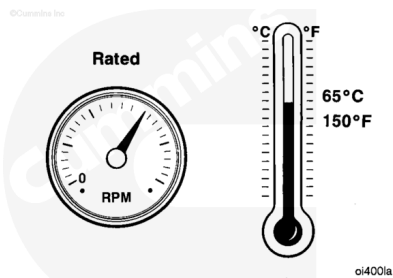
Remove the two plugs from the filter housing.

Install the end of the valve assembly in the port of the filter nearest to the centrifuge from where the plug has been removed.

Install the end of the valve assembly in the port of the filter furthest away from the centrifuge from where the other plug has been removed.



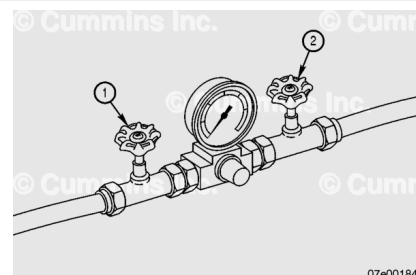
Operate the engine at rated rpm until the engine oil reaches its normal operating temperature. Continue running the engine at rated rpm for the duration of the pressure test.



Close valve (1), open valve (2), and record the pressure value before the Eliminator™ filter.

Close valve (2), open valve (1), and record the pressure value after the Eliminator™ filter.

If the difference in pressure is more than the maximum allowable value, inspect and clean the Eliminator™ filter screens before checking the oil pressure differential again.



Measurements			
		kpa	psi
Measurements	Maximum Oil Pressure Differential	60	9

Shut the engine OFF.

Remove the valve assembly from the Eliminator™ filter and install the two plugs.

Tighten the plugs.

Torque Value: 10 n•m [89 in-lb]

Where an oil sampling fitting is used, install the fitting in the position marked 1.

Tighten the oil sampling fitting.

Torque Value: 10 n•m [89 in-lb]

